

ECON 282E — Final Project Guidelines

Assemble: From Components to a Research Answer

Foundations of Macroeconomics
University of California, Riverside

Teams: Thursday, November 5 • Talks: Thursday, December 3 • Paper: Thursday, December 10, 2026

The idea. In this course you learn to be the **project manager** of a quantitative research project. You formalize the question, choose the model, define the equilibrium, pick the method, find the data, check quality, and communicate the answer. AI assistants write much of the code; the decisions and the validation are yours. The two homework assignments each build one *component* of that skill set. The final project **assembles** them, in a team, to answer a question the team poses itself.

Weight. 40% of the course grade: proposal 5%, talk and discussant role 15%, paper and replication package 20%.

1 The ladder: HW1 → HW2 → project

Each rung removes scaffolding and adds ownership.

	HW1: the representative-agent engine	HW2: heterogeneity and learning	Final project: assembly
Sessions	1–5	6–9	all; Session 10’s agentic workflow
Dates	out Oct 8 and 15; due Oct 29	out Oct 29 and Nov 12; due Mon Nov 23	teams Nov 5; proposal Nov 16; talk Dec 3; paper Dec 10
Who sets the question	the instructor (common spine), plus your own pitch (B5)	a shared core, plus a route question you choose	your team , evolved from the pitches
What you learn to manage	charter, unit test, data provenance, method choice, reproducibility	scoping a route, benchmarking an ML solver against a classical one, auditing someone else’s work	integration, division of labour, internal quality control, discussing another team’s work
Quality control	AI red-team item; validation ledger	AI red-team item; peer replication audit of a classmate’s HW1	internal audit; discussant report
What you carry forward	a calibrated RA model, a data pipeline, a method library, a pitch	an HA solver and one learning or measurement module	—

2 The integration rule

The team’s question must be answered by **combining at least one HW1 component and at least one HW2 component built by its members**, plus the piece the team adds: a new model feature, new data, or a new method. Examples of components:

- **From HW1:** the FRED data and calibration pipeline (A3); a validated global, perturbation or collocation solver (B2); an estimation routine (route R2); a capital-tax extension (route R3); a second-order solution (route R1).
- **From HW2:** the Aiyagari solver and stationary distribution (C1); the Euler-residual network (C2); an RL agent (C3); a text-based index (C4); a Krusell–Smith solver (route H1); a transition-path solver (route H2); a deep or RL solver for the HA household (route H3); a text- or exposure-disciplined income process (route H4).

Validate before you extend. Each reused component must first reproduce its own homework benchmark inside the team’s repository — the same numbers, from the team’s code, on the team’s machine — before it is extended. This is the course’s version of “replicate first, explore second”: the trusted benchmark comes before the new result. A project that assembles its components correctly and validates them rigorously, with a modest extension, earns a strong grade.

3 Required elements

Every project, whatever its emphasis, must have:

1. **A question the team formalized**, stated in one sentence, and labelled positive or normative. Say what number or ranking answers it.
2. **A model** from the Session 2 and Session 6 family (Aiyagari, Krusell–Smith, OLG, optimal policy), or a justified alternative. Say why a simpler model will not do, and define the equilibrium.
3. **One classical and one AI/ML method, cross-validated on at least one object** (a policy function, a price, a distribution, an index). Report where they disagree and why.
4. **Data with provenance**, structured (FRED, BLS, PSID, ...) or text, acquired by script, with series IDs or URLs and access dates. Public data only.
5. **An agentic workflow** (Session 10). The repository carries a `CLAUDE.md` (or the equivalent file for your agent) that describes the project to the next agent and the next human, and an `AI_LOG.md` that records the prompts that mattered, the decisions you made, where the AI was wrong and how you caught it, and an honest split of human and AI contribution.
6. **An internal audit**: each member re-runs a component built by another member and records the result with `templates/audit_report.md`.
7. **One-command replication**: `python run_all.py` (or equivalent) reproduces every number and figure from raw inputs. Fixed seeds; pinned environment; a `README` with run instructions. **No API keys in the repository** (use `.env` and `.gitignore`).

Never submit code you have not read. Every member must be able to explain every result in the paper.

4 Emphases

The syllabus lists four tracks. They are not separate projects but **emphases**: they say where the team's added piece lies. Each connects to homework routes whose components it naturally reuses.

Emphasis	The added piece	Natural components
A. A quantitative model and a policy experiment	a new friction, policy or counterfactual in an HA or OLG model, with welfare by group	HW1 R3; HW2 C1, H1, H2
B. A hard model solved with deep learning or RL	a learning solver where grids strain, validated against a classical benchmark	HW1 R1; HW2 C2, C3, H3
C. The economics of AI	AI as a shock to tasks, wages or capital, in a model you can solve	HW1 R2 or R3; HW2 C1, H4
D. Text or LLM measurement feeding a model	a validated text-based measure that disciplines a parameter or a shock	HW1 R2; HW2 C4, H4

Teams may mix emphases. A strong project usually crosses one line of this table: for example, a text-measured shock (D) fed into an Aiyagari policy experiment (A).

5 Milestones

Date	Milestone
Thu Oct 29	HW1 due, including your one-page pitch (B5). Post the pitch, under your name or anonymously, in the forum category <i>Pitches and teams</i> , and read the others.
Thu Nov 5	Teams (2–3) and topic , posted in <i>Pitches and teams</i> : members, the one-sentence question, and which members' pitches it grew from.
Mon Nov 16	Two-page proposal (5%; <code>templates/proposal.md</code>): the question; the HW1 and HW2 components you will reuse and their benchmarks; the gap your team fills; the division of labour; a risk register. Written feedback by Session 9 (Nov 19).
Mon Nov 23	HW2 due. Your HW2 components should now be ready to reuse.
Thu Dec 3, Block B	Talks and discussants (15%). Slots: with at most five teams, a 10-minute talk, 3 minutes from the discussant and 2 of Q&A; with six teams, 8 + 2 + 2. Each team discusses another team's project (<code>templates/discussant_form.md</code>); the pairing and the other team's slides are sent by Monday, November 30.
Thu Dec 10	Paper of at most 10 pages, replication package , <code>AI_LOG.md</code> , internal audit reports and the discussant form (20%).

6 Deliverables

1. **Paper** (at most 10 pages, references included): question, model and equilibrium, components reused and how each was validated, calibration and data provenance, method and its accuracy, results, limitations. Written like a short working paper. Every number in it must appear in the output of the replication command.
2. **Repository**: `CLAUDE.md`, `README`, pinned environment, fixed seeds, data-acquisition scripts (no manual downloads), a short data dictionary, and one-command reproduction.
3. `AI_LOG.md`, with a validation ledger linking each headline claim to the check that supports it, as in HW1.

4. **Internal audit reports**, one per member.
5. **Talk slides** and the **discussant form** for the team you discussed.

7 Rubric

Dimension	Points	What we look for
Framing and question	15	A precise, well-motivated question; the right model and equilibrium concept; why a simpler model will not do
Integration and correctness	20	Reused components reproduce their benchmarks; they are combined correctly; the added piece is right
Validation	20	Hand checks, benchmarks, a classical-vs-ML cross-validation, sanity checks; honest about what failed
Reproducibility and provenance	15	One-command reproduction on a clean machine; scripted data with access dates; pinned environment
AI-collaboration quality	10	An architect, not a consumer: an informative <code>AI_LOG.md</code> , a useful <code>CLAUDE.md</code> , honest attribution
Communication	10	A clear paper and talk; figures that tell the story
Discussant and audit	10	A specific, fair, well-evidenced discussant report; internal audits that actually re-ran the code

How the rubric maps to the course grade. The proposal (5%) is graded on the framing dimension and on the quality of the component plan and risk register. The talk and discussant role (15%) is graded on communication and on the discussant dimension. The paper and replication package (20%) are graded on the remaining dimensions. All members of a team receive the same project grade unless the audit reports or the `AI_LOG.md` show a clearly unequal contribution.

8 Coverage: where each session's tools are used

The two homework assignments and the project together exercise every key model and technique of the course. **Core** items are required of everyone; **route** items deepen one area by choice; the project **integrates** them.

Sess.	Key models and techniques	HW1	HW2	Project
1	growth model; planner and equilibrium; dynamic programming; why RL	A1, A2 (core)	C3	model choice
2	Bewley–Huggett–Aiyagari; Krusell–Smith; OLG; Ramsey and time consistency	A1(d) (core); R3	C1 (core); H1, H2	emphasis A, C
3	VFI; Euler equations and time iteration; Python, PyTorch; data workflow	A2, A3 (core)	C2	replication package
4	root-finding, optimization, interpolation, quadrature; Tauchen and Rouwenhorst	B1, B2 (core); R2	C1	calibration, estimation
5	perturbation, Blanchard–Kahn, pruning; Chebyshev collocation, Smolyak, finite elements	B2 (core); R1	C2 benchmark	emphasis B
6	Aiyagari as a nested fixed point; Krusell–Smith; OLG transitions; Negishi	—	C1 (core); H1, H2	emphasis A
7	neural networks; training; Euler-residual and deep VFI solvers; validation	—	C2 (core); H3	classical-vs-ML check
8	Q-learning, policy gradients, actor–critic; HA at the frontier; taxation without commitment	—	C3 (core); H1, H2, H3	emphasis B
9	text as data; embeddings; LLMs as instruments; retrieval	A3(c) (AI audit)	C4 (core); H4	emphasis C, D
10	agentic research workflows; verification; governance	AI red-team	AI red-team	CLAUDE.md, AI_LOG.md, audit

Ground rules. AI assistance is encouraged as a partner, not a ghostwriter. Public data only; no personal, IRB- or HIPAA-protected data. Cite every source and separate your contribution from prior work. Templates: `homeworks-exams/FinalProject/templates/`.